

Finite Configurations Cannot Generate a Constant Trace in Rule 30

David Lee Condrey
WritersLogic, Inc.
david@writerslogic.com

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We classify both constant central-trace fibers of Rule 30 after prescribing the right half of the initial configuration. When $R_0 = 0$, the all-zero trace forces either the zero row or an infinite alternating left tail selected by the first positive right-hand one. When $R_0 = 1$, the all-one trace forces the same infinite checkerboard left half for every right half. Consequently the zero row is the only finitely supported point of the zero fiber, and the one fiber contains no finitely supported point. No column in the orbit of a nonzero finite configuration is therefore eventually constant. If the support lies in $[-w, w]$, the largest last time through which the center can remain constant is $w + 1$; equivalently, the maximum constant-prefix length is $w + 2$. Exactly 2^w configurations attain this length when w is even, and $2^w - 1$ when w is odd.

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1 Introduction

Rule 30 is the binary cellular automaton on $\{0, 1\}^{\mathbb{Z}}$ whose local rule is the following.

$$F(x)_i = x_{i-1} \oplus (x_i \vee x_{i+1}) \tag{1}$$

It is left permutive: fixing x_i and x_{i+1} , the value $F(x)_i$ is a bijection of x_{i-1} .

Wolfram’s Prize Problem 1 asks whether the center column generated by the lone seed $\dots 0001000 \dots$ can ever become periodic [7]. Kopra records the broader width-one question for every nonzero finite configuration [4, after Thm. 3.1.12]. The results below settle the complete period-one case for this broader configuration class, and hence prove in particular that the lone seed’s center cannot become constant. They do not settle Prize Problem 1 and are not presented as a prize submission: every nonconstant eventual period remains open.

The mechanism is an exact inversion: prescribing the right half and demanding an all-zero center forces the left half cell by cell, as a prefix-OR of the right half. A right half whose first one sits at depth 3, for instance, forces $L_1, L_2, \dots = 0, 0, 1, 0, 1, 0, 1, \dots$, with ones at every odd depth past the third. The same one-sided bijectivity underlies Rowland’s unique-history construction [5, Prop. 1]. What makes the present classification Rule-30-specific is the OR in (1), which latches; the analogous claim is false for Rule 90 (Section 3). Figure 1 shows the forced left half and that contrast.

Jen proves that Rule 30 can have at most one eventually periodic temporal sequence under a finite initial condition [2, Thms. 2a, 2b, 3][1, Prop. 3]; Kopra’s modern formulation excludes periodic traces of width two but not width one [3, Cor. 3.7]. Jen also gives necessary and sufficient mechanisms for constant nonzero temporal sequences in the relevant class of nearest-neighbor rules [2, Thm. 7a]. We make no novelty claim for the bare all-one exclusion. What is not supplied by those results is the right-half-by-right-half classification of either constant fiber or the sharp finite-radius horizon and extremizer count proved here. For comparison, Voorhees solves a one-step predecessor problem for binomially determined nearest-neighbor *additive* automata over $\mathbb{Z}_p$ by an analogue of backward integration, while recording that the general predecessor problem is NP-complete [6]. Our inversion imposes a trace constraint over all future times on the nonadditive Rule 30.

2 The constant-trace fibers

The trace and the two halves of a row are written as follows.

$$c_t := \text{Tr}(x)_t = F^t(x)_0, \quad R_j := x_j, \quad L_j := x_{-j} \quad (j \geq 0) \quad (2)$$

Throughout, $\mathcal{F}$ denotes the finite-support configurations.

The trace sees only column 0, but under a left-permutive rule a discrepancy left of the origin propagates rightward along a diagonal and must eventually reach it. The classification rests on this one fact, which uses only locality and left permutivity, not the specific form of (1).

Lemma 1 (Triangular uniqueness). *If two configurations agree at every coordinate $i \geq 0$ and have the same central trace, then they are equal.*

Proof. If $-n$ is their rightmost differing coordinate, locality preserves agreement to its right through time n . Left permutivity preserves the discrepancy along $(-n, 0), (-n+1, 1), \dots, (0, n)$, contradicting equal traces. In its finite form, if the rows agree on $i \geq -r$ and their traces agree for $0 \leq t \leq r$, the same induction forces agreement at $-r, -r+1, \dots, -1$.

Uniqueness alone does not say what the forced left half looks like. The next result computes it: the rows with zero trace form one invariant class $\mathcal{C}_m$ for each position m of the first right-hand one, each class is carried to its predecessor by F , and the left half is a prefix-OR of the right. Figure 1(a) is a member of $\mathcal{C}_3$.

Theorem 2 (Zero-trace fiber). *For a prescribed right half with $R_0 = 0$, the zero-trace-compatible left half is unique. Suppose the positive half is nonzero, and let $m = \min\{j \geq 1 : R_j = 1\}$. The forced left half is then the following, with R_j for $j > m$ arbitrary.*

$$L_j = 0 \quad (j < m), \quad L_m = 1, \quad L_j = j \bmod 2 \quad (j > m) \quad (3)$$

If instead the positive half is zero, the unique row is zero. The same classification in prefix-OR form is the following.

$$L_{2k+1} = \bigvee_{j=1}^{2k+1} R_j \quad (k \geq 0) \quad (4)$$

$$L_{2k} = R_{2k} \wedge \neg \bigvee_{j=1}^{2k-1} R_j \quad (k \geq 1) \quad (5)$$

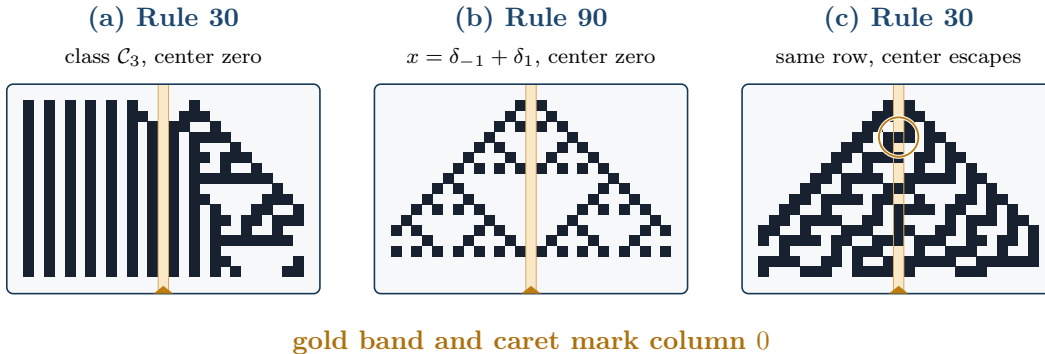


Figure 1: Time runs downward. In (a) the forced left tail alternates, hence is infinite, which is what Corollary 4 exploits. Panels (b) and (c) run one finite row under both rules; the difference is the OR latch in (1).

Proof. For $m \geq 1$, let $\mathcal{C}_m$ consist of rows satisfying these conditions. One Rule 30 step gives the three identities below.

$$\begin{aligned} x'_0 &= L_1 \oplus R_1 \\ R'_j &= R_{j-1} \oplus (R_j \vee R_{j+1}) \\ L'_j &= L_{j+1} \oplus (L_j \vee L_{j-1}) \end{aligned} \tag{6}$$

Suppose $m > 1$. These identities then give the following.

$$x'_0 = 0, \quad R'_j = L'_j = 0 \ (j < m - 1), \quad R'_{m-1} = L'_{m-1} = 1, \quad L'_m = m \bmod 2 \tag{7}$$

For $j > m$, including $j = m + 1$ because $L_m = 1$, the OR in the last update is one, so $L'_j = (1 - (j \bmod 2)) \oplus 1 = j \bmod 2$. Thus $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$. For $m = 1$, the alternating left half is fixed, $x'_0 = 1 \oplus 1 = 0$, and $R'_1 = 0 \oplus (1 \vee R_2) = 1$ independently of the right tail; hence $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$. Every $\mathcal{C}_m$ therefore realizes the zero trace.

A nonzero right half selects a realization by its first one, and Lemma 1 makes it unique; a zero right half forces a zero left half by the same lemma against the zero row.

The other constant trace has a simpler fiber: one row per right half, and the left half no longer depends on which right half it is.

Theorem 3 (All-one fiber). *For a prescribed right half with $R_0 = 1$, the left half compatible with an identically one trace is unique and is the same for every such right half.*

$$L_j = 1 \iff j \text{ is positive and even} \tag{8}$$

Proof. Write $L_0 := x_0 = R_0 = 1$, so that the third identity of (6), $L'_j = L_{j+1} \oplus (L_j \vee L_{j-1})$, applies at every $j \geq 1$. Under (8) it gives $0 \oplus (1 \vee 0) = 1$ for even j and $1 \oplus (0 \vee 1) = 0$ for odd j , so the left half is fixed by F whatever the right half is. At the origin, $F(x)_0 = L_1 \oplus (L_0 \vee R_1) = 0 \oplus 1 = 1$ independently of R_1 , so the trace is identically one and $L_0 = 1$ is restored at every step. Lemma 1 makes the left half unique.

3 Finite configurations

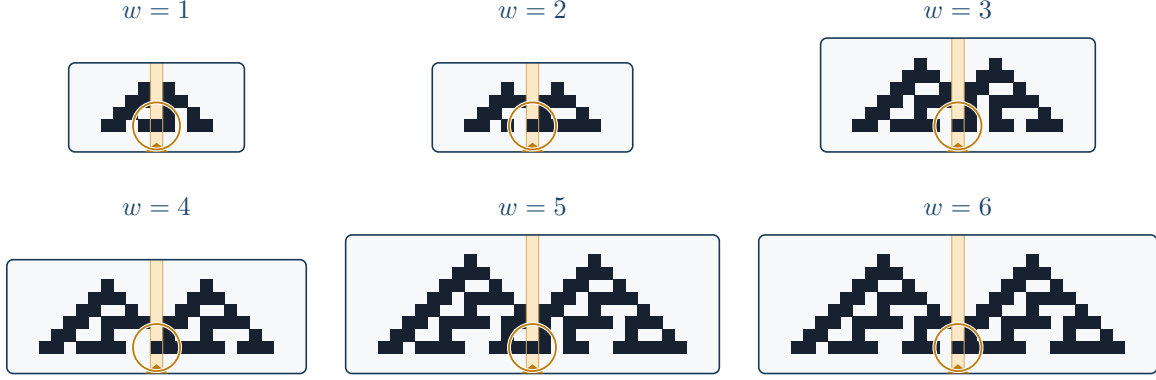
Every nonzero row in Theorem 2 carries ones arbitrarily far to the left, so none of them is finitely supported. That is the whole of the next statement, once one checks that a finite orbit cannot become the zero row.

Corollary 4. $\text{Tr}^{-1}(0^{\mathbb{N}}) \cap \mathcal{F} = \{0\}$. *Moreover, no column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually zero.*

Proof. Every nonzero row in Theorem 2 has ones at all sufficiently large odd left depths. If a column of a finite orbit were zero from time T onward, translation would make $F^T(x)$ a finite zero-trace row. It is nonzero: if $[a, b]$ is the support interval of a nonzero finite row, its next row has ones at $a - 1$ and $b + 1$ because Rule 30 maps 001 and 100 to one.

Corollary 4 is not a consequence of left permutivity alone: Rule 90 is also left permutive, yet the finite row with ones at ± 1 keeps a zero center forever (Figure 1(b,c)). The Rule-30-specific step is the OR latch $R'_1 = 1 \vee R_2 = 1$ in the proof of Theorem 2, which has no analogue in an additive rule.

The same latch upgrades the corollary from one constant to both, with no further machinery. Theorem 3 proves the second half again and independently, since (8) has ones at every positive even depth; the argument below is shorter and is the one to quote.



circled cell = first nonzero center, at time $2\lceil w/2 \rceil + 1$

Figure 2: One extremizer of Theorem 6 per radius, truncated from the zero-trace completion at left depth w . The horizon is flat across each pair $w = 2k - 1, 2k$, which is the parity effect in the statement.

Corollary 5. *No column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually constant.*

Proof. The eventually-zero case is Corollary 4. Suppose instead some column is eventually one; translating, take it to be column 0, so $c_t = 1$ for all $t \geq T$. Applying (1) at the origin gives $l_t = c_{t+1} \oplus (c_t \vee r_t)$, and $c_t = 1$ collapses the disjunction, so $l_t = 1 \oplus c_{t+1}$. For $t \geq T$ we also have $c_{t+1} = 1$, hence $l_t = 0$. Column -1 is then zero from time T onward, which Corollary 4 forbids.

4 A sharp finite horizon

Corollary 4 says a finite row's center cannot stay zero forever. How long it can stay zero, as a function of the support radius, is exactly one step past the light cone of the first odd depth beyond that radius.

For $w \geq 1$, let $\mathcal{S}_w = \{x \neq 0 : \text{supp}(x) \subseteq [-w, w], x_0 = 0\}$. For $x \in \mathcal{S}_w$, the zero-center horizon is defined as follows.

$$H(x) = \max\{h : F^t(x)_0 = 0 \text{ for } 0 \leq t \leq h\} \quad (9)$$

Theorem 6 (Sharp finite horizon). *Let $w \geq 1$. Then both of the following hold.*

$$\max_{x \in \mathcal{S}_w} H(x) = 2 \left\lceil \frac{w}{2} \right\rceil, \quad \#\left\{x \in \mathcal{S}_w : H(x) = 2 \left\lceil \frac{w}{2} \right\rceil\right\} = 2^w - 1 \quad (10)$$

Proof. Let q be the least odd integer greater than w , so $q - 1 = 2\lceil w/2 \rceil$. If the positive right half is nonzero, a trace zero through time q , compared with its zero-trace completion, would force by finite-prefix uniqueness and (4) $x_{-q} = \bigvee_{j=1}^q R_j = 1$, contradicting $q > w$. If that half is zero, comparison with the zero row makes every nonzero left cell visible by time w . Thus $H(x) \leq q - 1$.

Conversely, choose any of the $2^w - 1$ nonzero words $(R_1, \dots, R_w)$, install (4)–(5) through left depth w , and zero all other cells. This row agrees with its infinite zero-trace completion through the light cone of time $q - 1$; when w is odd, the extra depth $w + 1$ is even and forced to zero. The missing odd cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

The one-trace horizon is the same computation against (8), and the parities exchange roles: ones sit at even depths, so the binding depth is the least *even* one past the radius, and $R_0 = 1$ already makes the row nonzero, which is why no extremizer is lost here.

Theorem 7 (Sharp finite horizon, one). *Let $w \geq 1$, let $\mathcal{T}_w = \{x : \text{supp}(x) \subseteq [-w, w], x_0 = 1\}$, and for $x \in \mathcal{T}_w$ let $H_1(x) = \max\{h : F^t(x)_0 = 1 \text{ for } 0 \leq t \leq h\}$. Let q be the least even integer greater than w . Then both of the following hold.*

$$\max_{x \in \mathcal{T}_w} H_1(x) = q - 1, \quad \#\{x \in \mathcal{T}_w : H_1(x) = q - 1\} = 2^w \quad (11)$$

Proof. A trace one through time q , compared with the all-one completion of Theorem 3, would force by finite-prefix uniqueness and (8) $x_{-q} = 1$, contradicting $q > w$; thus $H_1(x) \leq q - 1$. Conversely, choose any of the 2^w words $(R_1, \dots, R_w)$, install (8) through left depth w , and zero all other cells. This row agrees with its all-one completion through the light cone of time $q - 1$; when w is even, the extra depth $w + 1$ is odd and forced to zero. The missing even cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

Combining the two horizons, the sharp law over both constants is flat in the parity that splits each one separately.

Corollary 8. *Let $w \geq 1$ and let $x \neq 0$ have $\text{supp}(x) \subseteq [-w, w]$. The longest constant prefix of the central trace of x has length at most $w + 2$, and exactly 2^w rows attain length $w + 2$ when w is even, while exactly $2^w - 1$ do so when w is odd.*

Proof. For $x_0 = 0$ Theorem 6 gives $2\lceil w/2 \rceil$, which is w for even w and $w + 1$ for odd w ; for $x_0 = 1$ Theorem 7 gives $q - 1$, which is $w + 1$ for even w and w for odd w . These quantities are the last matching time indices, so the corresponding prefix lengths are larger by one. The maximum last index is $w + 1$ in both parities, attained in the one case when w is even and in the zero case when w is odd, with the corresponding counts 2^w and $2^w - 1$.

5 Conclusion

Theorems 2 and 3 classify the two constant fibers of the trace map, and Corollaries 4 and 5 show that the zero configuration is the only finite point in the zero fiber and that the one fiber has no finite points; translating the observation column gives the exclusion for every column. Theorems 6 and 7 quantify the finite obstruction exactly. The next unresolved trace case is period two.

A Reproducibility and formalization

The audited computational source is repository commit [99b2d82af113e385f43452eb2b8f1b22180e6de0](#). The manuscript branch changes no file under `13-rule30/experiments/rule30`. From the repository root, the independent numerical audit is run by

```
uv run python 13-rule30/experiments/rule30/constant_trace_audit.py 7
```

It ends with **FAILURES: none** and returns the last matching time index and number of extremizers shown below. Prefix lengths are the displayed time indices plus one.

w	H_0	count	H_1	count	$\max H$	count
1	2	1	1	2	2	1
2	2	3	3	4	3	4
3	4	7	3	8	4	7
4	4	15	5	16	5	16
5	6	31	5	32	6	31
6	6	63	7	64	7	64
7	8	127	7	128	8	127

Every cell in Figures 1 and 2 is generated from (1). The first command checks the claims without writing; the second regenerates all nine TikZ fragments used by the two figures.

```
cd 13-rule30/experiments/rule30
uv run python figures.py --check
uv run python figures.py
```

The check prints

```
figures: all claimed properties hold
```

Regeneration begins its output with

```
figures: wrote TikZ fragments to
```

followed by the clone-local paper `figures` directory. Figure 1 checks an all-zero center for the displayed $\mathcal{C}_3$ row, an all-zero Rule-90 center for the row with ones at $\{-1, 1\}$, and escape of that same row under Rule 30. The six panels in Figure 2, for $w = 1, \dots, 6$, have first nonzero center times 3, 3, 5, 5, 7, 7. The numerical table above is the complete expected table from the independent audit.

The focused executable tests are run in the same directory by

```
uv run python -m unittest test_zero_tail_probe.py
```

The expected result is `Ran 11 tests` followed by `OK`. Finally, the PDF is built from `13-rule30/docs/rule30/paper` with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error zero-tail-note.tex
```

The accompanying `Rule30ZeroTail.lean` is a partial formalization. It formalizes the local equation and left permutivity, propagation of a rightmost discrepancy and triangular unit sensitivity, the prefix-OR closed form assuming a first positive right-hand one, the local Boolean identities for the $\mathcal{C}_m$ invariant, and the non-left-bounded conclusion conditional on the classification disjunction. It does *not* derive that disjunction from an all-zero trace, and it does not formalize the all-one fiber, the finite-support corollaries, or either sharp horizon and extremizer count. The formalization is therefore supporting verification, not a machine-checked proof of the main theorems.

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References

- [1] E. Jen, Aperiodicity in one-dimensional cellular automata, *Physica D: Nonlinear Phenomena* 45(1–3) (1990), 3–18. DOI: 10.1016/0167-2789(90)90169-P.
- [2] E. Jen, Global properties of cellular automata, *Journal of Statistical Physics* 43(1–2) (1986), 219–242. DOI: 10.1007/BF01010579.
- [3] J. Kopra, Rapid left expansivity, a commonality between Wolfram’s Rule 30 and powers of p/q , *Theoretical Computer Science* 946 (2023), 113668. DOI: 10.1016/j.tcs.2022.12.018.
- [4] J. Kopra, *Cellular Automata with Complicated Dynamics*, doctoral dissertation, University of Turku (2019). URN:ISBN:978-952-12-3891-8.
- [5] E. S. Rowland, Local nested structure in Rule 30, *Complex Systems* 16(3) (2006), 239–258. DOI: 10.25088/ComplexSystems.16.3.239.
- [6] B. Voorhees, Predecessor states for certain cellular automata evolutions, *Communications in Mathematical Physics* 117(3) (1988), 431–439. DOI: 10.1007/BF01223374.
- [7] S. Wolfram, Announcing the Rule 30 prizes, 2019. Stephen Wolfram Writings.